Minsuh (Ben) Cho

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EDUCATION

Johns Hopkins University, Baltimore, MD *Aug 2024 – Expected May 2030*
B.S. Mechanical Engineering & Applied Mathematics and Statistics
GPA: 3.76 / 4.0 *(Dean's List, Fall 2024 & Spring 2025)*

WORK EXPERIENCE

Research Assistant - Hopkins Extreme Materials Institute *May 2025 – Aug 2025*

- Designed experimental workflow and implemented a Gaussian Process Regression (GPR) optimization model for high-impact experiments, reducing laser flyer recast height by 35% and enhancing flyer integrity
- Manufactured Al and Cu samples by EDM and Polishing—producing controlled surface roughness samples for experiments
- CAD designed a compression grid to eliminate sample bulging within the holder assembly, improving alignment during laser shots
- Presented work results to 15+ members in lab group meeting

Mechanical Sub-Team - Mars Rover Team, JHU *Mar 2025 – Aug 2025*

- CAD and manufactured a parallel robotic gripper with custom-machined components, to achieve a secure grasp of up to 5 kg loads using a self-locking lead screw mechanism for enhanced grip stability
- Engineered a sheath auger drill capable of extracting soil from depths up to 10 cm, enabling reliable transfer of soil samples to the rover's spectroscopy analysis system
- Led design reviews for rover sub-systems, communicating design rationale from mission requirements and concept evaluation in motor selection, electronics, and prototyping process

Research Assistant - ICE-4-iMD Lab, JHU *Jan 2025 – May 2025*

- Developed a GPR model in Python to predict material strength, optimizing kernel selection for modeling dislocation behavior in metals
- Reviewed experimental datasets from literature for research on size-dependent dislocation plasticity

TECHNICAL SKILLS

- **Programming**: Python, Arduino(C++), MATLAB, HTML
- **CAD**: Solidworks, Creo
- **Manufacturing**: 3D Printing, CNC Machining(EDM, Lathe), Laser Cutting

PROJECTS

Co-Creator – Robotic Arm Launcher Arcade, JHU MechE Design II *Jan 2025 – April 2025*

- Developed an interactive arcade-style basketball game featuring a joystick-controlled robotic arm launcher, beam break scoring, and UI display system, using Arduino(C++)
- Led SolidWorks CAD design of the robotic arm launcher, hoop structures, and sensor housing; utilized 3D printing and laser cutting for rapid prototyping and assembly

Co-Creator – Glider Launcher Vehicle, JHU MechE Design I *Sep 2024 – Dec 2024*

- Led a three-member team to devise a vehicle that converts gravitational potential energy to elastic potential energy to launch a glider over a 6-ft away, 4-ft tall obstacle
- Utilized Solidworks CAD, laser-cutting, machining to design vehicle
- Awarded 2nd place in the MechE Design Competition at JHU against 22 teams

Creator – Demographics and Postural Data Correlation Analysis Capstone *Sep 2023 – May 2024*

- Conducted a correlation analysis of high school students between their demographic data(6 categories) and

postural behaviors to gain insights to help improve posture among high school teenagers
- Coordinated observational study sessions and analyzed +40 students' data using statistical methods such as Regression Analysis, Pearson's Correlation Coefficient, Point-Biserial Correlation, and Chi-Square Test

Creator – Posture Detection/Recommendation Web App *May 2023 – Nov 2023*
- Developed a web app that incorporates an open-source posture detection model to analyze the sitting posture of computer users and provide recommendations
- Built app features that provide data analytics of user's wrong posture data and custom-matched recommendations/stretch exercise videos for postural improvement
- Produced visual graphics of postural data for users to monitor their improvement

AWARDS

- Cum Laude Society 2024
- Bausch and Lomb Honorary Science Award 2023
- Colonel D.F. Wills Economics Award 2024
- AIME Qualifier 2022